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Brief Report

T2F: a domain-agnostic multi-agent framework for unstructured text to factuality evaluation items generation

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Received: 29 December 2024 / Accepted: 2 May 2025
Published online: 27 May 2025
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Abstract
Large language models (LLMs) demonstrate exceptional linguistic capabilities in text generation but remain prone to factual errors, particularly in specialized domains. Traditional factuality evaluation methods primarily rely on human annotation, which is costly, time-consuming, and difficult to generalize across different domains. To address these limitations, this study proposes an innovative multi-agent framework-T2F (Text-to-Factuality)-designed to automatically convert unstructured text into high-quality factuality evaluation datasets. T2F operates through the coordinated efforts of four specialized agents: Analysis, Information Extraction, Generation, and Validation. By systematically processing input data, T2F autonomously generates multiple types of assessment items-including single-choice questions, fill-in-the-blank questions, and true/false statements-without requiring human annotation, while maintaining strong cross-domain applicability. Experimental results demonstrate that T2F achieves data conversion success rates of 99% in the World Heritage domain, 98% in the Medical domain, and 85% in the Film domain. The generated data effectively assess LLMs' factuality accuracy, highlighting T2F's capability as a scalable and domain-agnostic factuality evaluation framework.

Keywords Multi-agent · Factuality evaluation · Large language models

1 Introduction

With the rapid development of Large language models (LLMs), these models have demonstrated remarkable generative capabilities in the field of natural language processing, and are now widely applied across various domains such as education [1], healthcare [2, 3], law [4], and finance [5]. However, the quality of content generated by these models is often questioned, particularly regarding its factual accuracy. Factuality evaluation, which measures the accuracy and reliability of a model's knowledge, has become a crucial research focus for ensuring the safety and usability of these models. Without effective evaluation of the factuality of model-generated content, its deployment could lead to serious consequences, including the spread of misinformation, misleading user decisions, and ethical concerns.

Existing factuality evaluation methods face multiple challenges. Most benchmark datasets rely heavily on human annotation, which is a labor-intensive and time-consuming process, hindering large-scale expansion across multiple

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domains. Additionally, these benchmarks exhibit limited flexibility and scalability, failing to provide high-quality factuality evaluation datasets tailored to specialized fields.

Although automated data synthesis methods offer a potential solution for generating test data, they still present limitations in interpretability, model generalization, and scalability. For instance, single-model approaches often operate as black boxes, making it difficult to trace decision-making logic. Moreover, they are constrained by context window limitations and computational resources, which hinder their ability to handle complex tasks. To address these challenges, a more effective strategy is to structure the factuality evaluation workflow and decompose it into multiple collaborative agents, thereby enhancing transparency, robustness, and flexibility, while improving data generation quality and adaptability.

To tackle these challenges, we propose T2F (Text-to-Factuality), a multi-agent framework for generating factuality evaluation datasets. Rather than serving as a fully optimized, final solution, T2F provides a structured workflow that explores how to automatically generate factuality evaluation datasets from unstructured text. By leveraging task decomposition and agent collaboration, T2F aims to improve the transparency, controllability, and scalability of the data generation process, laying the foundation for future optimizations and improvements.

Specifically, this paper makes the following three key contributions:

(1) *A Generalized Framework for Factuality Evaluation Data Generation*: T2F does not rely on fixed tools or predefined domains. By incorporating a domain-adaptive analysis mechanism, the framework can be directly applied to multiple domains without modification, enhancing the universality and flexibility of the data generation process.
(2) *Well-Designed Task Decomposition*: T2F systematically decomposes the factuality evaluation data generation process into four distinct modules: text analysis, information extraction, question generation, and quality verification. This modular design not only improves the transparency of the data generation pipeline but also ensures low coupling between components, facilitating future extensions, customization, and module replacement.
(3) *Validation Mechanism Balancing Effectiveness and Efficiency*: T2F employs a validation paradigm centered on the accuracy of the final output samples, effectively mitigating the negative impact of error accumulation across pipeline stages. Additionally, this design avoids excessive verification overhead in terms of time and cost, achieving a balance between maintaining high data quality and improving overall data generation efficiency.

Experiments conducted on datasets from three domains-disease and symptoms, movie plots, and heritage lists-demonstrate that T2F can generate high-quality factuality evaluation data and effectively assess mainstream LLMs.

2 Related work

The factuality evaluation of LLMs is a critical step in ensuring the accuracy and reliability of their generated content. Through systematic evaluation, shortcomings in the models' factual performance can be identified, providing guidance for subsequent improvements. Evaluation benchmarks, metrics, and methodologies constitute the cornerstone of factuality assessment.

2.1 Factuality evaluation benchmarks

Benchmarking evaluates the factuality of LLMs from two primary perspectives. On one hand, it assesses the models' ability to accurately identify factual information within a given context, typically employing objective evaluation methods such as single-choice and true/false questions. On the other hand, it examines the factual consistency of model-generated content, which is closely tied to text generation tasks. From a domain perspective, mainstream benchmarks predominantly focus on assessing the general capabilities of LLMs, encompassing diverse problem types such as text generation, question answering, language understanding, and reasoning. Representative works in this area include HaluEval [6], FreshQA [7], FACTOR [8], and Pinocchio [9]. Additionally, a growing number of benchmarks are beginning to target domain-specific applications, such as those in medicine [10], journalism [11], and history [12].

2.2 Factuality evaluation methods

In terms of evaluation metrics and methodologies, human assessment remains the simplest and most effective approach for evaluating the factuality of LLMs. However, in practice, human evaluation faces significant limitations, including time-consuming processes, high costs, and potential biases introduced by subjective judgments of evaluators. To address these challenges, researchers have increasingly explored and developed automated evaluation methods, which can be broadly categorized into three types:

Statistical Computation-Based Methods: For objective questions such as single-choice or true/false, metrics like accuracy and F1-Score are commonly employed. For subjective questions, metrics such as BLEU (Bilingual Evaluation Understudy) [13], ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [14], and Perplexity can quantify the overlap between the outputs of LLMs and reference sentences, indirectly assessing factual alignment. Dong [15] proposed the KaRR method, which evaluates the factual capabilities of a model by calculating the ratio of the probability of generating the correct knowledge to the probability of generating the answer randomly, offering a more targeted approach.
Model-Based Evaluation Methods: Several methods leveraging pre-trained language models have demonstrated strong performance in generating textual representations and assessing textual similarity. These approaches can be used to compare the differences between LLM-generated content and ground truth. Representative works include BERTScore [16], BLEURT [17], and BARTScore [18]. Additionally, methods such as GPTScore [19], GPT-judge [20], and CONNER [21] attempt to directly apply LLMs in factuality evaluation tasks. These methods have proven effective not only in judging factuality but also in evaluating the relevance and utility of information, providing users with richer and more actionable feedback.

Comprehensive Evaluation Frameworks: With the widespread adoption of LLMs across diverse application scenarios, single evaluation methods often fall short in comprehensively assessing the outputs of such models. To address this limitation, researchers have proposed a range of integrated evaluation frameworks [26–28]. These frameworks combine multiple evaluation techniques to verify factuality from different perspectives, balancing the high precision of human evaluation with the efficiency of automated methods.

2.3 Methods for synthesizing factuality test data

Although existing evaluation benchmarks and methods have made notable progress in improving the factuality of LLMs, they primarily rely on manually annotated datasets, resulting in high costs and limited efficiency. Against this backdrop, the automated construction of test datasets for factuality evaluation is emerging as a key research focus.

Utama [22] proposed a data generation pipeline, Falsesum, which leverages a controllable text generation model to perturb human-annotated summaries, thereby producing datasets with diverse factual errors. Soleimani [23] introduced NonFactS, a data generation model based on the BART framework, capable of synthesizing non-factual summaries given a contextual document and a factually accurate summary. Rykov [24] utilized GPT-4 and LLaMA-2 models to generate factual data and demonstrated its effectiveness in training hallucination detection models. Additionally, Scir [25] proposed an end-to-end factuality evaluation approach, LLM-Oasis, which generates factual and non-factual texts by extracting statements from Wikipedia and introducing fabrications to support evaluation tasks.

A comparative analysis of the proposed method and existing approaches is presented in the Table 1. T2F exhibits a higher degree of automation, requiring minimal human intervention while synthesizing high-quality factuality evaluation data. Furthermore, it demonstrates strong generalization capabilities, enabling direct application across different domains without modification.

Discover Artificial Intelligence (2025) 5:77 | https://doi.org/10.1007/s44163-025-00294-w

Table 1 Comparison between existing data generation methods and T2F

Method	Functionality	Workflow	Automation level	Generation type
Falsesum [22]	Generate NLI datasets for evaluating factual consistency in summaries.	Use rule-based methods to distort factual information in text and store it in a structured format, then train a T5 model to generate summaries with factual errors.	The model primarily generates negative samples, while positive samples still require manual annotation. The T5 model needs to be retrained when switching tasks or domains.	Capable of generating both intrinsic and extrinsic factual error texts.
NonFactS [23]	Generate non-factual summary data to train factuality evaluation models.	First, construct a dataset using rule-based methods, then train a BART model on this dataset to generate non-factual information.	Constructing the training set requires manually designed rules, and the BART model needs to be retrained for cross-domain scenarios.	Non-factual text.
Rykov [24]	Generate non-factual data to train hallucination detection models.	Directly generate data using prompt engineering with GPT-4 and LoRA-tuned LLaMA2-7B, then filter low-quality samples using rule-based methods.	Manual data annotation is required when fine-tuning the LLaMA2-7B model for data generation.	Factual and non-factual text pairs.
LLM-Oasis [25]	Generate factuality verification data in an end-to-end manner.	Guide the LLM through prompt engineering to perform claim extraction, claim fabrication, and factuality test text generation.	Rely on manual review and data annotation for sample validation.	Factual and non-factual text pairs.
T2F (ours)	Convert unstructured text into factuality evaluation datasets.	Introduce a multi-agent framework to decompose the factuality evaluation data generation task into sample analysis, information extraction, question generation, and verification stages.	Highly automated with no manual annotation required. Applicable to multiple domains without modification.	Capable of generating single-choice, fill-in-the-blank, and true/false questions.

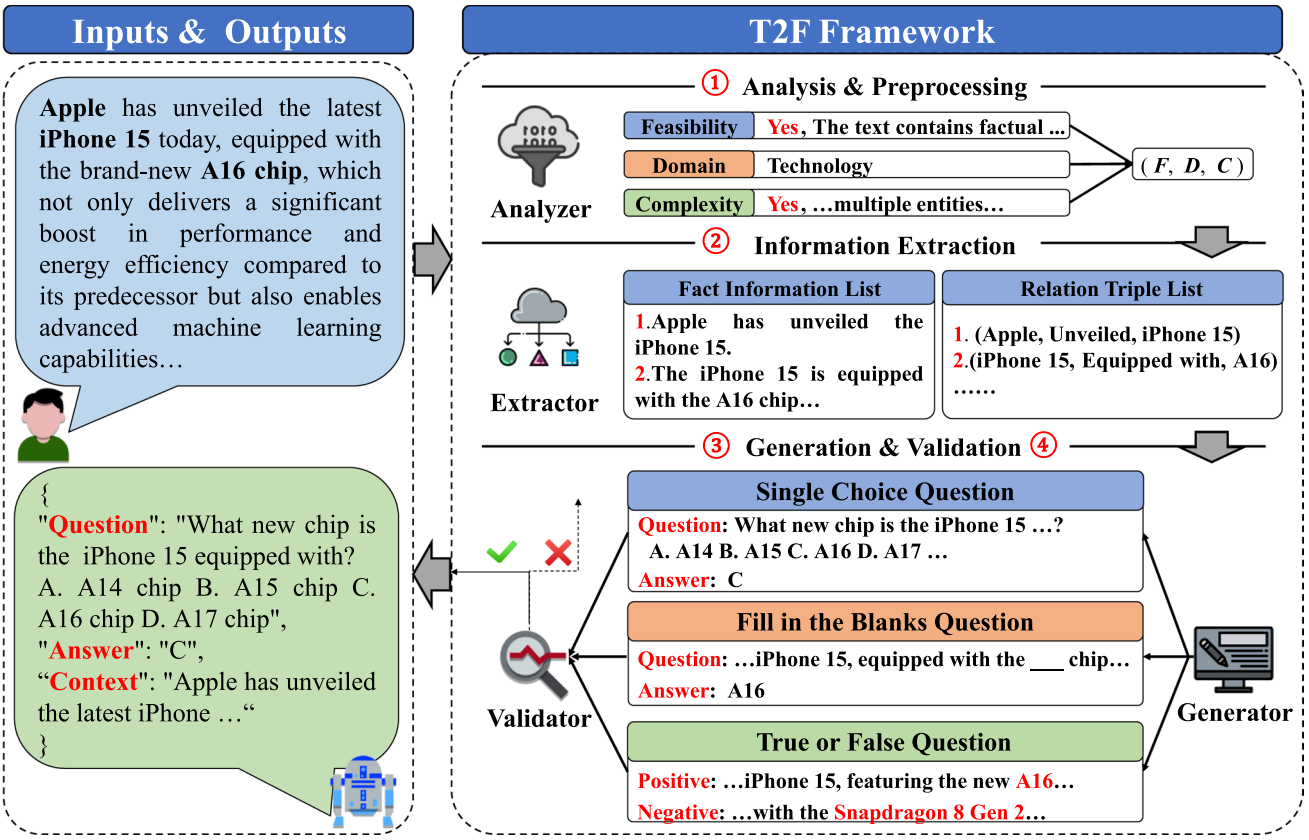


Fig. 1 An Overview of the T2F Framework

3 Method

3.1 Overview

The T2F framework is designed to transform unstructured text into factuality evaluation datasets through a multi-agent system, whose overall structure is illustrated in Fig. 1. It operates as a pipeline where each agent performs a specialized task, as shown in Algorithm 1, collectively ensuring the generation of high-quality evaluation items. The process begins by analyzing the input text to determine its feasibility for transformation and identifying its domain. Based on the complexity of the content, additional processing is performed to extract key relationships or contextual information. This structured data is then used to generate factuality evaluation items, including questions and answers, which are subsequently validated for accuracy and relevance. By modularizing the workflow, T2F ensures scalability, domain adaptability, and automation throughout the dataset creation process.

Algorithm 1 A detailed outline of the T2F pipeline

Input: Unstructured Text Dataset $T = \{t_1, t_2, \ldots, t_n\}$, Agents $\mathcal{A} = \{\mathcal{A}_A, \mathcal{A}_E, \mathcal{A}_G, \mathcal{A}_V\}$
Output: Validated Factuality Evaluation QA Dataset QA_{final}
$QA_{final} \leftarrow \emptyset$
for each $t_i \in T$ **do**
  $(D_i, C_i, F_i) \leftarrow \mathcal{A}_A(t_i)$ ▷ Analyze raw text
  if $\neg F_i$ **then**
    continue ▷ Skip non-factual texts
  end if
  $\mathcal{L}_i \leftarrow \mathcal{A}_E.\text{ExtractFacts}(t_i, D_i)$ ▷ Extract factual lists
  $\mathcal{R}_i \leftarrow \emptyset$
  if C_i **then**
    $\mathcal{R}_i \leftarrow \mathcal{A}_E.\text{ExtractRelations}(t_i, D_i)$ ▷ Extract relation triple
  end if
  $(Q_i, A_i)^{FBQ} \leftarrow \mathcal{A}_G.\text{GenerateFBQ}(t_i, \mathcal{L}_i, \mathcal{R}_i)$
  $(Q_i, A_i)^{SCQ} \leftarrow \mathcal{A}_G.\text{GenerateSCQ}(t_i, \mathcal{L}_i, \mathcal{R}_i)$
  $(Q_i, A_i)^{pos}, (Q_i, A_i)^{neg} \leftarrow \mathcal{A}_G.\text{GenerateTFQPairs}(t_i, \mathcal{L}_i, \mathcal{R}_i)$
  $QA_i \leftarrow (Q_i, A_i)^{FBQ} \cup (Q_i, A_i)^{SCQ} \cup (Q_i, A_i)^{pos} \cup (Q_i, A_i)^{neg}$
  for each $(Q_i, A_i) \in QA_i$ **do**
    if $\mathcal{A}_V((Q_i, A_i), t_i, \mathcal{L}_i)$ **then** ▷ Verify factual consistency
      $QA_{final} \leftarrow QA_{final} \cup \{(Q_i, A_i)\}$
    end if
  end for
end for
return QA_{final}

3.2 Analysis agent

The analysis agent is the first module in the T2F framework, responsible for performing an initial analysis of the input unstructured text. This agent, driven by LLMs, carries out three core tasks: feasibility analysis, domain analysis, and complexity analysis.

The objective of feasibility analysis is to assess whether the text contains verifiable factual information. Using carefully designed prompt templates, the analysis agent directs the LLM to evaluate whether the text represents a clear fact. If the text is deemed factual, subsequent processing steps are performed; otherwise, it is discarded.

Domain analysis is employed to identify the domain type of the given text, enabling the domain information to be passed to subsequent fact extraction and complexity analysis modules via prompting. This strategy allows the model to leverage its domain-specific knowledge, facilitating more effective key information extraction and ultimately improving the overall quality of the evaluation dataset.

In the complexity analysis, the agent provides a predefined set of complex relationships S to the LLMs to assist the model in identifying intricate relationships within the text. This set encompasses various types of potential complex relationships, such as multi-step reasoning, relational chains between multiple entities, conditional logic, and temporal sequences. The LLMs, guided by these prompts, analyze whether the input text involves any of these complex entity relationships or reasoning processes. Specifically, the analysis can be expressed using the Eq. 1, where $c_i(T)$ indicates whether the text T contains the i -th complex relationship from the set S, and $\mathbb{I}$ is the indicator function, taking the value of 1 when $c_i(T)$ is true, and 0 otherwise. Ultimately, if the text T includes at least one complex relationship, then $C(T) = 1$, indicating that the text is complex; otherwise, $C(T) = 0$, indicating that the text is relatively simple.

$$C(T) = \mathbb{I}\left(\sum_{i=1}^{S} \mathbb{I}(c_i(T)) > 0\right) \tag{1}$$

3.3 Information extraction agent

The primary tasks of this agent include fact extraction and relation extraction, aimed at extracting structured factual information from text and identifying complex relationships between entities, thereby providing high-quality data for subsequent question generation. The task of fact extraction involves identifying and extracting explicit factual information from unstructured text. Facts typically consist of specific entities (such as persons, locations, and events) along with attributes or relations that describe these entities.

For more complex texts, the information extraction agent analyzes the entities and their relationships, outputting them in the form of (entity, relation, entity) triplets, as described in Eq. 2, where $\mathrm{E}(T)$ denotes the extraction of entities from the raw text, and $\mathrm{IR}(e_i, e_j, T)$ represents the inference of the relationship between entities e_i and e_j within the text T.

$$\begin{aligned} R = &\{(e_i, r_i, e_j) \mid (e_i, e_j) \in \mathrm{E}(T), \\ &r_i = \mathrm{IR}(e_i, e_j, T)\} \end{aligned} \tag{2}$$

However, in open-domain scenarios, the entity types present in the input text are often uncertain, making it challenging to achieve fully automated identification and extraction using existing information extraction tools. To enhance the generality of T2F, this agent employs an LLM as its kernel, enabling end-to-end information extraction without requiring manual intervention. Moreover, benefiting from T2F's modular design, this agent can be substituted with mature information extraction tools when applied to well-defined domains, ensuring adaptability to different application scenarios.

3.4 Generation agent

The question generation agent is a core module responsible for generating factual evaluation questions. It synthesizes the input text, domain information, factual data, and relational triplets (if available) to generate questions. The tested LLMs cannot be definitively categorized as either open-source or closed-source. Therefore, to enhance the generalizability of T2F, the generation agent selects single-choice questions (SCQ), true/false questions (TFQ), and Fill-in-the-blank questions (FBQ) as generation targets. This choice enables subsequent analysis of factuality issues in LLMs using straightforward metrics such as accuracy. The specific details are as follows:

(1) SCQ: Each question consists of one correct option and several distractors.
(2) TFQ: These assess whether a statement derived from the text is factually correct, with the agent generating both a positive and a negative sample.
(3) FBQ: These focus on core facts and entities by masking key information in the input text.

The agent employs a one-shot learning approach to ensure the generation of high-quality questions. The underlying principle is described by Eq. 3, where L denotes the fact information list, $P_{\mathcal{G}}$ represents the prompt templates for different question types, and $U_{\mathcal{G}}$ refers to the user-specified question type. Overall, the prompt template used for generation agent is shown in Fig. 2.

$$(q_k, a_k) = \mathcal{G}(T, L, R; P_{\mathcal{G}}, U_{\mathcal{G}}) \tag{3}$$

3.5 Validation agent

Since T2F primarily processes unannotated, unstructured data and is inherently constrained by the capabilities of its underlying LLM, errors may arise in various stages such as fact extraction and relation extraction. While fine-grained factuality detection can help identify potential error sources within individual modules, integrating independent verification mechanisms at each stage in T2F's multi-module system would significantly increase computational costs and system complexity.

To address this, T2F adopts a design paradigm centered on ensuring the accuracy of the final output samples. Specifically, we employ a validation agent to comprehensively verify the generated questions and answers. By comparing the

The prompt template.

Here is an example illustrating how to generate a single-choice question for factual consistency evaluation, to test the subject's ability to identify factual alignment:
[Sample original text] {Demo text}
[Generated factual consistency evaluation question] {Demo question}
[Answer] {Demo answer}
Now, based on the user's input of the original text and the extracted facts, generate one single-choice factual consistency evaluation question:
[Original text] {Input text}
[Extracted facts] {Facts}
Note: You may also refer to the extracted list of (entity, relationship, entity) triples to ensure the validity of the question and enhance the plausibility of incorrect options.
[Triples list] (e, r, e)
Please output the result in JSON format as follows:
{"Question": Include the question and candidate options A, B, C, D, "Answer": The correct option}

Fig. 2 The prompt template used by generation agent

generated content against key information in the original text, the validation agent retains only those samples that are confirmed to be error-free, thereby ensuring the overall quality of the dataset. This approach allows the system to tolerate certain errors in earlier processing stages while guaranteeing the accuracy and reliability of the final output through rigorous post-generation filtering. The underlying principle is represented by the Eq. 4. Q_{final} denotes the set of valid samples.

$$Q_{\text{final}} = \{(q_k, a_k) \mid \mathcal{V}(q_k, a_k, T, L) = 1\} \tag{4}$$

4 Experiments

4.1 Setup

4.1.1 Examined LLMs

We employed GPT-4o as the core engine for data analysis, generation, and validation tasks. The evaluation of the tested models was conducted from two perspectives:

(1) *Closed-Source Models*: We assessed the performance of representative closed-source models on synthetic data, including GPT−3.5, GPT-4o [29], Gemini-Pro [30], and Baichuan4.
(2) *Open-Source Models*: We evaluated the performance of open-source models, focusing on LLaMA-3-Instruct (70B and 8B versions) [31], Qwen [32], and the latest version of DeepSeek-v3 [33].

The fundamental details of these LLMs are summarized in Table 2. In the experiments, all LLMs were used in their non-quantized versions, and the temperature parameter was set close to zero to mitigate the impact of output randomness.

4.1.2 Datasets

To validate the applicability and effectiveness of our proposed method across different domains and text structures, we selected three publicly available datasets. These datasets span multiple domains, including some that have been underexplored in the context of factuality evaluation research. The datasets are described as follows:

Table 2 Details of baseline LLMs to be evaluated

Model	Organization	Size (#P)	URL	OS
GPT-3.5,GPT-4o	OpenAI	~	https://chat.openai.com	×
Gemini-pro	Google	~	https://gemini.google.com	×
LLaMA-3-Instruct	Meta	8B,70B	https://llama.meta.com	✓
DeepSeek-v3	DeepSeek	671B(MoE)	www.deepseek.com	✓
Baichuan4	Baichuan Inc.	~	www.baichuan-ai.com	×
iFLYTEK Spark-v3.5	iFLYTEK	~	https://xinghuo.xfyun.cn/	×
Qwen2.5-Instruct	Alibaba	72B	https://tongyi.aliyun.com/qianwen	✓

It lists the model's name, parameter size (indicated as "~" to denote undisclosed specifics), URL, and their open-source (OS) status

(1) *Disease & Symptom Dataset*: This dataset describes medical conditions and their associated symptoms, along with recommended treatments.
(2) *Movie Plot Dataset*: This dataset provides story summaries for various movies, along with related information such as directors and release dates.
(3) *World Heritage List*: This dataset is sourced from UNESCO's World Heritage List and includes descriptive information about ancient sites or historical buildings.

We extracted 100 samples from each dataset for experimental analysis. For the Disease and Symptom and Movie Plot datasets, we employed various templates to reorganize the key information into unstructured narrative text. Additionally, we visualized the text length distribution across the three datasets, as shown in the Fig. 3. Compared to the other two datasets, the Movie Reviews dataset exhibits a longer average text length.

4.1.3 Evaluation metrics

For validity verification, we used accuracy as the primary metric. For SCQ and TFQ, we developed additional scripts to extract and refine the answers from the model's responses. For FBQ, LLMs often generate additional redundant content-such as detailed analytical processes or further explanations-instead of providing a concise answer. Therefore, we adopted a more lenient evaluation criterion for these questions. Specifically, a model's output is considered correct if it contains the key answer keywords, regardless of any extra content. This approach avoids penalizing correct responses that include supplementary information and better reflects the model's understanding.

To evaluate the effectiveness of data generation, we introduced multiple metrics, including:

(1) Rare Word Ratio (RW Ratio): The proportion of rare words in the generated data.
(2) Similarity: For SCQ, we computed the similarity between different answer choices. For TFQ, we measured the similarity between positive and negative samples generated from the same data.

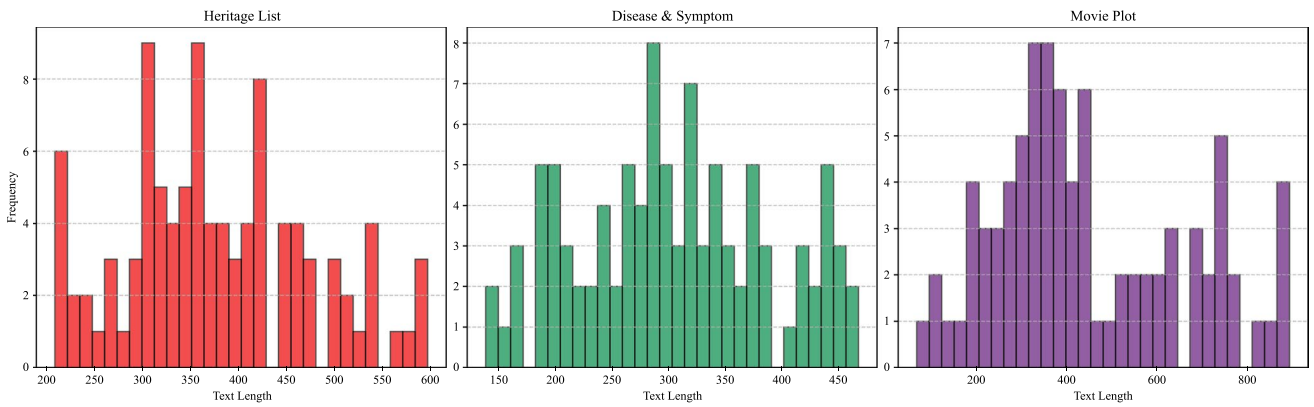


Fig. 3 Text length distribution across the three datasets

(3) Average Accuracy (Avg-Acc): We evaluated the dataset using GPT-4o and GPT−3.5 models, leveraging their average accuracy to reflect the dataset's difficulty and novelty.
(4) Error Count (# Errors): The number of generated samples containing erroneous information detected by the validation agent.

4.2 Results and analysis

4.2.1 Validity verification

First, we analyze the effectiveness of T2F, with the results presented in Table 3. When processing datasets from three distinct domains-Heritage List, Disease & Symptom and Movie Plot-T2F achieves a data conversion success rates of 99%, 98%, and 85%, respectively.

In Table 3, #Total denotes the total number of samples in the original dataset. #Feasible refers to the number of samples identified by T2F as containing factual information. #Complex represents the subset of samples that T2F determines to involve intricate logical relationships or require additional reasoning for proper handling. Lastly, #Errors indicates the number of question-answer pairs that T2F detects as containing erroneous information during the verification phase.

The experimental results demonstrate that T2F effectively generates a large number of valid samples for inherently factual domains such as medical and geographical datasets. In contrast, the movie plot dataset-despite containing explicit factual details like director and release date-exhibits a lower number of feasible samples due to its inherent diversity and complexity, especially in genres like science fiction or pure fiction. This outcome is primarily attributed to T2F's feasibility analysis module, which acts as a safeguard against hallucinations and logical inconsistencies by filtering out texts that fall outside the model's reliable processing capabilities. In doing so, the module enforces a conservative strategy that, while potentially over-filtering in some domains, guarantees robustness and reliability across datasets. Furthermore, the comprehensive input information enables the model to generate nearly error-free samples that pass validation, confirming that T2F meets user requirements even under diverse domain conditions.

Next, to assess whether T2F can be effectively applied in evaluation scenarios, we conducted an evaluation of nine LLMs in a closed-book setting, where models were unable to reference the original text while generating responses. The evaluation results are presented in Table 4.

The test data generated by T2F effectively differentiates the factuality performance of mainstream LLMs. The models exhibit a clear performance gradient across different question types: stronger models (e.g., GPT-4o) consistently achieve higher accuracy across all tasks compared to weaker models (e.g., GPT−3.5). This suggests that the generated evaluation questions achieve the intended level of discriminative power, successfully avoiding two common pitfalls: (1) overly simple questions, which could lead to artificially high scores across all models, and (2) poorly designed questions, which could result in uniformly low performance across models. Consequently, the T2F benchmark ensures both effectiveness and reliability in evaluating LLM factuality.

From the perspective of question type, FBQ tasks are more challenging than the other two categories. This is primarily because model responses are not constrained by a predefined candidate answer set, making LLMs more prone to generating factually incorrect outputs. These findings further indicate that in open-ended tasks, where LLMs cannot reference the original text, their generative nature makes them more susceptible to factual inconsistencies.

Additionally, Fig. 4 presents examples of three types of questions generated by T2F for the disease & symptom dataset. These examples demonstrate that the framework can effectively transform raw text into factuality evaluation questions while providing accurate reference answers to facilitate verification.

Table 3 The quality of samples generated by T2F based on three datasets

	Heritage List	Disease & Symptom	Movie Plot
# Total	100	100	100
# Feasible	99	98	85
# Complex	22	15	33
# Errors	0	0	0

Table 4 Evaluation results of mainstream LLMs on T2F-synthesized data

Dataset	Models	FBQ	TFQ	SCQ
Heritage List	GPT-3.5	0.6061	0.8737	0.7677
	GPT-4o	0.7172	**0.9646**	0.8788
	LLaMA-3-70B-Instruct	0.6970	0.9293	0.8788
	LLaMA-3-8B-Instruct	0.5354	0.8687	0.7677
	Gemini-pro	0.6566	0.9242	0.7374
	DeepSeek-v3	**0.7475**	0.9293	**0.8990**
	Qwen2.5-72B-Instruct	0.6869	0.9596	0.8485
	iFLYTEK Spark-v3.5	0.5556	0.7727	0.8081
	Baichuan4	0.6061	0.9141	0.7879
Disease & Symptom	GPT-3.5	0.6633	0.9184	0.9184
	GPT-4o	0.7755	**0.9745**	**0.9592**
	LLaMA-3-70B-Instruct	0.6939	0.9541	**0.9592**
	LLaMA-3-8B-Instruct	0.6122	0.9337	0.8673
	Gemini-pro	0.6939	0.9388	0.9286
	DeepSeek-v3	**0.8469**	0.9694	0.9388
	Qwen2.5-72B-Instruct	0.7041	0.9643	0.9286
	iFLYTEK Spark-v3.5	0.5918	0.8622	0.9286
	Baichuan4	0.7449	0.9490	0.8878
Movie Plot	GPT-3.5	0.3294	0.6353	0.6824
	GPT-4o	**0.5412**	**0.8824**	**0.8824**
	LLaMA-3-70B-Instruct	0.4000	0.8176	0.7059
	LLaMA-3-8B-Instruct	0.2235	0.6353	0.7412
	Gemini-pro	0.2588	0.6471	0.5882
	DeepSeek-v3	0.4941	0.5824	0.8588
	Qwen2.5-72B-Instruct	0.2706	0.6235	0.7882
	iFLYTEK Spark-v3.5	0.2471	0.5588	0.5765
	Baichuan4	0.3647	0.5235	0.7647

The best-performing results are emphasized in bold

4.2.2 Performance comparison

To validate the effectiveness of the T2F framework design, we conducted a comparative analysis between T2F and a data generation approach that directly utilizes GPT-4o. Additionally, we examined the performance of T2F when its core LLM was replaced with GPT−3.5. The results, as presented in Table 5 and 6, indicate that T2F-generated samples exhibit superior quality across multiple dimensions, specifically in the following aspects:

(1) Rich Entity Information: T2F-generated samples contain a higher proportion of rare words, indirectly reflecting a greater presence of entity-rich content, such as geographical locations, personal names, and domain-specific terminology.
(2) Higher Distractor Strength: In SCQ and TFQ, the answer choices generated by T2F demonstrate greater similarity, thereby increasing the difficulty and distractor strength of the questions.
(3) Stronger Discriminative Power: In most cases, LLM performance on T2F-generated data is lower, suggesting that T2F samples are more challenging and contain novel information, making them more effective in differentiating model capabilities.

The experiments also highlight the critical role of the core LLM in T2F. While employing a weaker LLM may lead to erroneous sample generation, our findings demonstrate that even when using GPT−3.5 as the core model, T2F still produces higher-quality samples than those generated directly by GPT-4o.

Input Text

A diagnosis of Cellulitis is suggested by the presence of symptoms like Redness, Pain, tenderness, Swelling, Skin changes, Lymph node enlargement. Common management strategies involve Antibiotics, Warm compresses, immobilization, rest, Pain management.

FBQ

A diagnosis of Cellulitis is suggested by the presence of symptoms like Redness, Pain, tenderness, Swelling, Skin changes, and ___ node enlargement.
[Correct Answer]: Lymph

SCQ

Which of the following is a common management strategy for Cellulitis?
A. Surgery B. Antibiotics C. Chemotherapy D. Radiation therapy
[Correct Answer]: B

TFQ

[Positive] A diagnosis of Cellulitis is indicated by symptoms such as redness, pain, tenderness, swelling, skin changes, and lymph node enlargement. Common management strategies include antibiotics, warm compresses, immobilization, rest, and pain management.
[Negative] A diagnosis of Cellulitis is suggested by the presence of symptoms like Redness, Pain, tenderness, Swelling, Skin changes, Lymph node enlargement. Common management strategies involve Antiviral medication, Cold compresses, increased physical activity, and dietary changes.

Fig. 4 Sample instances generated by T2F on the Disease & Symptom dataset

Table 5 Performance comparison of T2F and unoptimized direct GPT-4o data generation in SCQ synthesis

Datasets	Methods	RW Ratio↑	Similarity↑	Avg-Acc↓	# Errors↓
Heritage List	GPT-4o	0.2210	0.0464	0.8674	1
	$T2F_{GPT\text{-}3.5}$	0.1933	**0.0720**	0.9022	5
	$T2F_{GPT\text{-}4o}$	**0.2363**	0.0703	**0.8233**	**0**
Disease & Symptom	GPT-4o	0.1086	0.0188	0.9541	**0**
	$T2F_{GPT\text{-}3.5}$	**0.1551**	**0.0244**	**0.9000**	8
	$T2F_{GPT\text{-}4o}$	0.1437	0.0242	0.9388	**0**
Movie Plot	GPT-4o	0.2380	0.0355	**0.7083**	1
	$T2F_{GPT\text{-}3.5}$	0.2626	**0.0487**	0.7911	6
	$T2F_{GPT\text{-}4o}$	**0.3301**	0.0404	0.7824	**0**

The best-performing results are emphasized in bold

↑ indicates that the larger the metric, the better the effect, while ↓ indicates that the smaller the metric, the better the effect. When evaluating the baseline models, we exclusively use samples that have passed the feasibility analysis

4.2.3 Robustness analysis

Since the T2F framework adopts a pipeline structure, there exists a certain degree of dependency between its agents, which may lead to error propagation-errors introduced during the analysis or information extraction stages could potentially affect the subsequent sample generation process. To assess the robustness of T2F, we deliberately injected fabricated

Table 6 Performance comparison of T2F and unoptimized direct GPT-4o data generation in TFQ synthesis

Datasets	Methods	RW Ratio↑	Similarity↑	Avg-Acc↓	# Errors↓
Heritage List	GPT-4o	0.1366	0.6578	**0.9167**	**0**
	$T2F_{GPT\text{-}3.5}$	0.1208	0.5745	0.9256	10
	$T2F_{GPT\text{-}4o}$	**0.1379**	**0.6866**	0.9192	**0**
Disease & Symptom	GPT-4o	0.1142	0.5370	0.9490	**0**
	$T2F_{GPT\text{-}3.5}$	0.1077	0.5459	**0.9313**	7
	$T2F_{GPT\text{-}4o}$	**0.1155**	**0.5624**	0.9465	**0**
Movie Plot	GPT-4o	0.1238	0.6114	0.7929	1
	$T2F_{GPT\text{-}3.5}$	0.1150	0.5278	0.7879	5
	$T2F_{GPT\text{-}4o}$	**0.1274**	**0.6134**	**0.7589**	**0**

The best-performing results are emphasized in bold

Table 7 Effectiveness analysis of the T2F verification module

Methods	Heritage list	Disease and symptom	Movie plot
Strong verifier	5/99	3/98	6/85
T2F verifier	5/99	2/98	6/85

The values in the table represent the number of detected erroneous samples over the total number of samples (error samples/total samples). For the experiments, we exclusively use samples that have passed the feasibility analysis. The experiments were conducted using the SCQ samples

information-constructed based on the input context-into the fact list within the SCQ dataset to simulate adversarial perturbations. We then observed whether T2F's verification mechanism could effectively detect and filter out these erroneous entries.

To further compare verification performance, we implemented a strong verifier based on GPT-4o. Unlike T2F's built-in verification module, this verifier was explicitly provided with the injected false information to enhance detection effectiveness.

The experimental results, presented in Table 7, demonstrate that even under adversarial perturbations, the majority of T2F-generated samples remained factually accurate. Moreover, T2F's verification mechanism exhibited performance comparable to the strong verifier in identifying invalid samples, effectively filtering out the majority of erroneous data. These findings indicate that T2F demonstrates strong robustness and fault tolerance.

5 Limitations

Despite the promising potential of the T2F framework in converting unstructured textual data into datasets for factuality evaluation of LLMs, it still exhibits several limitations that warrant further investigation and improvement:

(1) *Potential Risks Due to Insufficient Text Content Filtering*: T2F does not implement rigorous content filtering when processing input text. While this design enhances the system's applicability and flexibility, it introduces potential risks of misuse by malicious actors to generate harmful or inappropriate content. Integrating more robust content moderation mechanisms is necessary to mitigate the likelihood of such abuse.
(2) *Limitations of LLMs' Reasoning Capabilities in Information Extraction*: The T2F framework relies on LLMs to perform information extraction tasks. However, the current reasoning capabilities of LLMs may fall short when handling complex or implicit information, resulting in incomplete data extraction. This incompleteness can directly impact the quality and reliability of the generated samples in subsequent processes.
(3) *Impact of Input Text Complexity on the Quality of Generated Questions*: During the process of generating questions based on input text, T2F may not consistently capture the most salient points, especially when processing lengthy texts with intricate details or generating fill-in-the-blank questions. Such deviations from the core content may lead to suboptimal question quality, characterized by weak relevance to the input text's main themes or insufficient depth.

(4) *Lack of Fine-Grained Evaluation Samples*: Currently, the T2F framework mainly generates three question types: FBQ, SCQ, and TFQ. This enables the evaluation of LLMs' factuality using coarse-grained metrics such as accuracy. However, this design limits fine-grained evaluation, as it lacks the ability to generate assessment items that reveal model performance at a more detailed level, particularly regarding specific information.

6 Conclusion

This paper presents T2F, a multi-agent framework for automating the generation of factuality evaluation datasets from unstructured text. By integrating specialized agents, T2F enables the scalable creation of high-quality evaluation items without manual annotation. Experimental results demonstrate its effectiveness in generating diverse factuality evaluation datasets across multiple domains, making it a valuable tool for testing the factual reliability of LLMs.

In the future, T2F could be further enhanced by integrating neuro-symbolic reasoning methods [34] and external knowledge graphs [35, 36] to improve the accuracy and depth of the generated evaluation items. Additionally, incorporating privacy-preserving mechanisms [37] would enhance its security. These advancements are expected to enable the framework to handle more complex reasoning tasks and expand its applicability to multilingual, highly specialized, and sensitive or privacy-critical domains, thereby advancing factuality evaluation capabilities in natural language processing.

Author contributions The authors confirm contribution to the paper as follows: study conception and design: Xin Tong, Bo Jin; data collection: Yasen Aizezi; analysis and interpretation of results: Xin Tong, Jingya Wang, Hanming Zhai; draft manuscript preparation: Xin Tong, Yasen Aizezi, Bo Jin. All authors reviewed the results and approved the final version of the manuscript.

Funding This work has been supported by the Natural Science Foundation of Xinjiang Uygur Autonomous Region, P.R. China, under the project "Research on Secure Electronic Data Access Scheme for Data Sharing" (2024D01A55) and the Fundamental Research Funds for the Central Universities (2024JKF13).

Data availability We utilized publicly accessible datasets: The Disease and Symptoms Dataset, accessible via www.kaggle.com/datasets/aadyasingh55/disease-and-symptoms. The Movie Plots Dataset, accessible via www.kaggle.com/datasets/ujwalkandi/unesco-world-heritage-sites. The UNESCO World Heritage List Dataset, accessible via www.kaggle.com/datasets/jrobischon/wikipedia-movie-plots.

Materials availability Not applicable.

Code availability The code can be obtained by submitting a request to the corresponding authors' email.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The author has no relevant financial or non-financial interests to disclose.

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